

Helium Producer Investment Analysis and Geopolitical Impact

Executive summary

Helium is a **strategic, non-renewable** industrial commodity with **no substitute in ultra-low-temperature cryogenics** (below roughly **-256 °C**, i.e., **-429 °F**). USGS emphasizes that nothing substitutes for helium in cryogenic applications requiring these temperatures, while some substitution is possible in welding, lifting gas, and certain diving mixes. ¹

The market's defining feature is **supply concentration**. Using USGS world production data (2023), the **United States (incl. withdrawals from Cliffside)** and **Qatar** together represent the overwhelming majority of global production, with Russia, Algeria, Canada, and a handful of smaller producers comprising the remainder. ²

Pricing is **not transparently quoted** like oil or copper. Government and industry observers note helium transactions are commonly governed by **confidential, bilateral contracts**; the USGS publishes an estimated "base price" for private Grade-A helium, and also reports surcharges were common during tight periods. ³

Over the last five full years (2021–2025), the **USGS estimated base price** for Grade-A helium rose from **\$7.57/m³ (2021)** to **\$11/m³ (2022)** and **\$14/m³ (2023–2024)**, then eased to **\$12/m³ (2025)**—still materially above 2021 levels. ⁴

Geopolitically, the helium supply chain is currently exposed to **Persian Gulf shock risk** through two coupled channels:

Upstream coupling to LNG: Qatar helium is produced from natural gas/LNG value chains; shutdowns at Ras Laffan therefore directly threaten helium output and contract fulfillment. ⁵

Maritime chokepoint risk: UNCTAD reports ship transits through the **Strait of Hormuz** have "come to a near halt," with **~97% drop** versus pre-conflict averages—disrupting logistics, insurance, and freight economics. ⁶

For investors, this implies helium exposure splits into two "styles":

Defensive / quality via the global industrial gas oligopoly (Linde, Air Liquide, Air Products), where helium is a small revenue line but a strategic product inside long-term contracts and high switching costs. ⁷

High beta / supply-scarcity optionality via smaller listed helium-focused producers/developers (e.g., Renergen, Royal Helium, Desert Mountain Energy, Pulsar), which can be more sensitive to helium tightness but carry **execution, financing, and concentration risk**. ⁸

Global supply-demand overview

Production sources by country and supplier concentration

USGS world production estimates (million m³ of helium, unless otherwise specified) show a market dominated by a few jurisdictions. In **2023**, “world total (rounded)” production is estimated at **~176 million m³**, with **Qatar (66)** and the **United States (64 extracted from natural gas + 18 from Cliffside Field)** the largest contributors. ⁹

In **2024e**, USGS estimates world production at **~180 million m³**, with Qatar (64), U.S. (68 extracted + 13 from Cliffside), and Russia (17) as the key supply pillars. ⁹

USGS also reports **2025e world production ~190 million m³**, alongside notable supply-side developments (new operations and storage), though the 2026 table’s U.S. country line appears internally inconsistent versus the reported world total; investors should treat 2025 country splits with caution and rely on the 2024e reconciled table for allocation work. ¹⁰

A practical interpretation for portfolio risk is that helium is **not only concentrated by country**, but also by a handful of mega-facilities and hubs (e.g., U.S. Wyoming/LaBarge–Shute Creek ecosystem, Qatar Ras Laffan, Russia Amur), which can create disproportionate price impact during outages. ¹¹

Estimated world helium production share (2023, USGS)

Recent capacity and operational changes

USGS highlights that in **2025**:

Six new U.S. helium operations began producing (three in New Mexico; one each in Colorado, Kansas, Montana).

¹²

A **helium storage cavern** was brought online in **Beaumont, Texas**, able to store “excess helium production.” ¹²

A new helium facility began operations in Canada, and another in South Africa.* ¹²

These changes matter because incremental plants and storage do not “fix” concentration risk, but they can reduce the amplitude of shortages when paired with stable logistics.

On the Russia side, restart and ramp-up of the **Amur Gas Processing Plant** has been a recurring swing factor; reporting indicates helium shipments from Amur began from at least one train in 2023, with additional ramps planned. ¹³

Inventories, storage, and the post-Federal Helium Reserve era

The U.S. Federal helium system transition remains central to understanding “inventory buffers”:

USGS reported remaining **Federal conservation helium inventory** at **~85.7 million m³** as of **9/30/2021**, and also documented private helium remaining in storage at Cliffside Field. ¹⁴

By **FY2022**, USGS reported remaining conservation helium **~60.7 million m³** and expected disposition of assets. ¹⁵

By **FY2023**, USGS reported remaining conservation helium **~51.5 million m³**, with the Federal Helium System's management transferred from BLM to GSA for disposal, and a court decision allowing the sale process to continue. ¹⁶

In **2024**, USGS reports the Federal Helium System assets were sold in two lots and transferred to one company on **June 27, 2024**. ¹⁷

Investment relevance: the U.S. market's former "strategic buffer" has been shrinking and shifting from a public-stockpile paradigm to a more commercial, contract-and-storage paradigm, increasing sensitivity to **plant outages** and **logistics shocks**. ¹⁸

Demand and end-use structure

USGS provides a clear end-use allocation for helium inside U.S. apparent consumption. In 2025, helium was used (in decreasing quantity) for: analytical/engineering/lab/science/specialty gases (22%); controlled atmospheres, fiber optics & semiconductors (17%); lifting gas (17%); MRI (15%); aerospace (9%); welding (8%); diving (5%); leak detection (5%); other (2%). ¹⁹

From a demand-elasticity standpoint, the most important segment is **healthcare and critical research** (MRI and labs) where switching is limited and conservation focuses on recycling and "closed-loop" systems rather than true substitution. USGS notes closed-loop recycling is becoming more common. ¹²

Price trends over the last five years

The USGS estimated "base price" series (private Grade-A helium) captures the market's tightening-and-relief pattern:

USGS estimated Grade-A helium base price (2021–2025)

USGS reported: **\$7.57/m³ (2021)**, **\$11/m³ (2022)**, **\$14/m³ (2023)**, **\$14/m³ (2024)**, **\$12/m³ (2025)**, often with surcharges. ⁴

Interpretation: even after the 2025 easing, the implied pricing regime remains elevated versus 2021, consistent with a market that has experienced repeated "shortage cycles." USITC documents **four notable helium shortages** (Shortage I–IV) in the past ~20 years. ²⁰

Public companies involved in helium

Helium exposure in public equities falls into three layers: integrated industrial gas majors (distribution + contracting), helium-focused producers (upstream + processing), and explorers/developers (optionality).

Comparable table of selected public companies

Notes on comparability:

Financials are a mix of US GAAP and IFRS; “EBITDA” may be adjusted or proxied depending on disclosure.

Helium revenue is rarely separated for industrial gas majors; where not disclosed, it is listed as “Not broken out” and treated qualitatively. ²¹

Company	Ticker(s)	HQ / primary listing	Role in helium value chain	Primary helium-linked assets / footprint	Helium production volume	Helium revenue disclosure	Latest revenue / profitability	Debt / leverage snapshot
Linde plc	LIN (Nasdaq)	UK/IE (global)	Industrial gases major (production, purification, distribution; long-term onsite contracts)	Global industrial gases footprint; electronics/ healthcare end markets; helium supplied within “rare gases” portfolio ²²	Not disclosed	Not broken out	2025 sales ~\$34.0B (reported) ²³	Illustrative EV/EBITDA data suggests debt/EBITDA context (aggregated) ²⁴
Air Products and Chemicals	APD (NYSE)	USA	Industrial gases major; meaningful U.S. helium refining/ distribution exposure	Large industrial gas portfolio; helium supplied into electronics/ healthcare; exposed to helium allocation/ shortage cycles ²⁸	Not disclosed	Not broken out	FY2025 GAAP included operating loss and special items (project exits/ actions) ²⁹	Aggregated metrics show ~\$18.3B and negative ³⁰

Company	Ticker(s)	HQ / primary listing	Role in helium value chain	Primary helium-linked assets / footprint	Helium production volume	Helium revenue disclosure	Latest revenue / profitability	Debt / leverage snapshot
Air Liquide (ADR proxy)	AIQUY (OTC ADR); AI (Euronext)	France	Industrial gases major (distribution + onsite supply); helium distributed as part of “rare gases”	Europe/US/Asia industrial gas network; major electronics and healthcare exposure 33	Not disclosed	Not broken out	2025: reported sales €26.94B and recurring operating income ~€5.58B , margin ~20.7% 34	Not consolidated here (by rep basis)
Renergen	REN (JSE)	South Africa	Helium + LNG producer (upstream wells + liquefaction)	Virginia Gas Project; produces LNG and liquid helium; Phase 1 and planned Phase 2 expansion 36	Phase 1 stated target ~350 kg/day liquid helium ; Phase 2 expansion adds ~4,200 kg/day (in addition to Phase 1) 37	More directly linked to helium/ LNG sales; commercial liquid helium sales announced 38	Early-stage ramp; production stability issues and restart noted 39	Project financial risk (no summ here)
Royal Helium	RHC (Canada)	Canada (TSXV)	Helium producer / purifier (midstream + upstream-linked)	Steeville purification facility (Alberta); ramping throughput and runtime 41	Operational updates: stable production at ~50% nameplate with periodic increases up to ~75% ; ~4 trailers/month at 50% 42	Revenue linked to helium sales (small, ramping)	Early commercial ramp	High cap financial risk

Company	Ticker(s)	HQ / primary listing	Role in helium value chain	Primary helium-linked assets / footprint	Helium production volume	Helium revenue disclosure	Latest revenue / profitability	Debt / leverage snapshot
Desert Mountain Energy	DME (TSXV); DMEHF (OTC)	Canada/ USA ops	Helium producer (processing + tube trailer delivery)	U.S. Southwest helium production and processing; commercial production announced ⁴⁴	"Commercial volumes" separated; deliveries via tube trailers (volumes not quantified publicly) ⁴⁴	More direct helium revenue linkage	Ramp phase; aiming to cover corporate G&A from production revenue ⁴⁵	Small-balance sheet
Pulsar Helium	PLSR (AIM/ TSXV)	UK/ Canada listing	Helium developer/ explorer	Topaz Project drilling program; delineation stage ⁴⁶	Not producing	N/A	Quarterly operating updates focus on drilling progress and financing ⁴⁷	Exploration stage dilution ⁴⁷

Shortlist for purchase consideration

This section frames "purchase suitability" as a function of **(a) desired risk profile**, **(b) need for direct helium beta**, and **(c) tolerance for geopolitical tail risk**.

Linde

Thesis: A high-quality way to own "helium scarcity" as part of a resilient industrial gas platform with durable pricing structures. Linde reported **~\$34B 2025 sales** and sustained profitability, and the sector is characterized by long contracts and local delivery economics (less exposed to tariffs, but still exposed to feedstock/logistics shocks). ⁴⁸

Helium linkage: Helium is not the majority driver, but helium tightness tends to strengthen pricing and customer stickiness across rare gases and specialty supply relationships; this is most valuable during shortage cycles documented by USITC and USGS. ⁴⁹

Valuation reality check: Aggregated valuation snapshots place Linde at a **high EV/EBITDA multiple (example snapshot ~18x)**—a "quality premium" that can compress in risk-off regimes. ²⁴

Catalysts to monitor: prolonged 2026 helium tightness (see Iran section), further long-term electronics/healthcare contracting, and supply re-optimization via storage/logistics. ⁵⁰

Risks: valuation compression; global industrial slowdown; rare-gas supply constraints (Ukraine/Russia related) and shipping disruptions impacting container availability. ⁵¹

Air Liquide

Thesis: Similar “defensive oxygen” profile with strong operating margins; 2025 results reporting indicates sales around **€26.94B** and recurring operating income **~€5.58B** with margin **~20.7%**. ³⁴

Helium linkage: Helium is embedded inside the specialty/rare gases offering, with demand tied to semiconductors and healthcare—segments that USGS identifies as major helium end uses. ¹⁹

Catalysts: Europe’s re-routing and sanctions-driven trade flow changes can raise the value of diversified supply portfolios; EU restrictions include an import ban on Russian helium effective September 2024, changing how Europe sources marginal volumes. ⁵²

Risks: FX and regional industrial cycles; less direct helium upside than pure plays.

Air Products

Thesis: Potential turnaround / re-rating candidate if capital discipline improves, but fundamentals include significant recent earnings noise: Air Products reported FY2025 GAAP results including **operating loss** and large business/asset action charges, versus FY2024 which included a large gain on its LNG business sale. ²⁹

Helium linkage: as a major U.S. industrial gas company, it is structurally exposed to helium supply tightness through customer relationships and specialty gas channels (not typically separated as helium revenue). ²⁸

Valuation snapshot: third-party aggregations show EV/EBITDA around low-teens (example ~12x), and significant net debt, but these metrics can swing with “special items.” ³⁰

Catalysts: improved project execution and reduced capex intensity; shortage-driven pricing power if helium remains tight.

Risks: execution and impairment risk; project portfolio reshaping; equity underperformance risk if macro weakens (recent 12-month price performance has been negative). ⁵³

Renergen

Thesis: A relatively rare listed name with more direct **liquid helium production linkage**. Renergen reports Phase 1 production/restart and has communicated Phase 1 output at **~350 kg/day liquid helium**, with Phase 2 expected to materially expand (including **4,200 kg/day** incremental liquid helium output in addition to Phase 1). ⁵⁴

Catalysts: successful stable operations (uptime), Phase 2 financing/commissioning, and persistent global helium tightness.

Risks: single-asset concentration; commissioning/ramp setbacks (the company itself describes production interruptions and restart timelines); sovereign/currency risk. ³⁹

Position sizing scenarios

These are **illustrative portfolio-weight ranges** for a diversified investor (not personalized financial advice). They reflect helium's volatility and the fact that public equities provide **indirect** exposure.

Conservative: 1–3% total “helium theme” exposure, primarily through one industrial gas major (e.g., 1–2% LIN or AIQY/AI) plus $\leq 0.25\%$ in one speculative helium producer if desired.

Moderate: 3–6% total, split across two industrial gas majors (2–4% combined) plus 0.5–1.5% across one or two helium-focused small caps (e.g., Renergen and/or a Canadian helium producer), recognizing dilution risk.

Aggressive: 6–10% total, with 3–6% in majors and 2–4% across multiple small-cap helium names; only appropriate if you accept that operational setbacks and financing cycles can dominate returns in the small-cap sleeve. The historical pattern of repeated shortage cycles underscores why this sleeve can be “boom-bust.” ⁵⁵

Iran conflict scenario analysis

Why Iran conflict matters disproportionately for helium

USITC previously highlighted that geopolitical conflict has repeatedly stressed helium supply, including the **2017 blockade of Qatar**, which cut off a large fraction of global supply and primed the market for shortages. ²⁰

In **March 2026**, the risk channel is clearer and more severe:

- **Qatar supply is structurally central.** USGS estimates Qatar produced **~66 million m³ in 2023** out of **~176 million m³** global output (~high-30% share). ⁹
- Industry reporting indicates QatarEnergy's Ras Laffan halt and force majeure have increased the likelihood of shortages and surcharges, with restart requiring **weeks** even after conditions allow resumption. ⁵⁶
- The Strait of Hormuz is a critical chokepoint for global trade; UNCTAD reports **ship transits dropping ~97%** versus pre-conflict averages, and that the strait carries substantial shares of global LNG and other seaborne trade. ⁵⁷
- The American Chemical Society's news outlet reports the war has taken **about one-third of the world's helium supply off the market**, and warns that if the conflict persists beyond **~two weeks**, disruptions could take **months** to resolve. ⁵⁸

Supply disruption scenarios

The table below expresses a probability-weighted view (probabilities are judgmental, not statistical forecasts). “Price impact” refers to directional pressure on contract renegotiations and spot/expedite costs; helium prices often have lags and opacity. ⁵⁹

Scenario	Description	Expected supply/ logistics effect	Time-to- normalization	Probability (illustrative)	Likely market impact
Short disruption	Limited outage, partial shipping normalization while security improves	Temporary Qatar helium/LNG interruption; expedited sourcing from U.S./Canada/ Algeria; container dislocation but manageable	1–2 months, given restart and ramp time even after resumption ⁵⁶	30%	Surcharges, priority allocations for critical users
Medium, constrained corridor	Qatar remains curtailed; Hormuz intermittently impaired (insurance + war risk keep traffic low)	Persistent logistics shock; “near halt” shipping effect constrains helium ISO container circulation ⁶⁰	2–6 months; ACS suggests months if >2 weeks ⁵⁸	45%	Broad allocations, spot/ expedite prices spike, higher working capital needs
Prolonged conflict / extended closure	Ras Laffan and corridor constraints persist	Structural shortage with multi-quarter impact; downstream rationing in semiconductors and labs	6–12+ months	25%	Severe supply rationing; contract repricing; capex acceleration for non-Gulf projects

Investor implications under Iran conflict

Industrial gas majors: likely to experience **mixed effects**—higher pricing power and contract leverage, but also potential volume constraints and customer force majeure disputes. Their diversified networks and long-term contracting can cushion earnings, but the risk is “supply unavailable,” not merely “price higher.” ⁶¹

Helium-focused producers outside the Gulf: could see **positive optionality** if they can deliver marginal volumes into constrained markets, but only if they have ready logistics (tube trailers, purification, export

pathways) and if customers can qualify supply quickly. Operational ramp-ups (Royal Helium, Desert Mountain) become especially valuable in this environment. ⁶²

Supply shocks timeline context

USITC documents four major helium shortage periods since 2005, driven by a combination of plant outages, geopolitical disruptions, and logistics constraints. ⁶³

Timeline of major helium supply shocks (illustrative)

Regulatory, technology, and demand-side drivers

Regulation and sanctions

EU measures expanded import bans to include **helium**, with a wind-down period to **26 September 2024**, and USGS notes the EU sanctions package was adopted June 2024 and effective September 2024. ⁶⁴

In practice, sanctions reshape **trade flows**, increasing the value of flexible, non-sanctioned supply and logistics, and can tighten regional markets even when global supply is adequate. USGS notes sanctions continued into 2025. ¹²

Technology and substitution

USGS is explicit: **no substitute for helium** in applications requiring cryogenic temperatures below about -256 °C; however substitution exists at the margin in welding (argon), lifting (hydrogen), and some diving mixtures. ⁶⁵

Technological mitigation is therefore primarily about **conservation and recycling**. USGS reports helium recycling is becoming more common and “closed-loop recycling systems” are increasing. ¹²

Demand-side factors by end market

MRI and healthcare: a major helium demand pillar; USGS estimates MRI accounted for **~15%** of U.S. helium uses in 2025. Supply disruptions disproportionately affect hospitals and research institutions that require stable cryogen supply. ⁶⁶

Semiconductors and controlled atmospheres: USGS places “controlled atmospheres, fiber optics, and semiconductors” at **~17%** of helium use in 2025—large enough that cyclical electronics demand can change market tightness. ⁶⁷

Space/aerospace: USGS lists aerospace pressurizing/purging at **~9%** in 2025. These uses tend to be highly schedule-driven, intensifying price spikes during shortages. ¹⁹

Supply chain vulnerabilities flowchart

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flowchart TD
    A[Natural gas fields with helium content] --> B[Gas processing / helium extraction]
    B --> C[Crude helium purification to Grade-A]
    C --> D[Liquefaction + ISO container filling]
    D --> E[Port export + shipping lanes]
    E --> F[Regional distribution hubs]
    F --> G[End users: MRI, semiconductors, aerospace, labs]

    subgraph Chokepoints & failure modes
        B1[Plant outages / feedgas issues]:::risk
        D1[ISO container scarcity / turnaround delays]:::risk
        E1[Strait chokepoints & war-risk insurance]:::risk
        F1[Allocation regimes / contract force majeure]:::risk
    end

    B -. -> B1
    D -. -> D1
    E -. -> E1
    F -. -> F1

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Risk mapping to current conflict: UNCTAD documents Hormuz disruption and near halt in ship transits, while industry reporting ties Ras Laffan shutdowns and force majeure to helium production interruptions.

68

Recent reporting on the Iran/Hormuz shock

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